

CASE STUDY QUESTIONS WITH ANSWERS

Data Frames in R Programming

Name: Savitha R

Reg no: 192421368

1. Case Study 1: Student Performance Analysis

Case Study: A university stores student information in a data frame containing Student_ID, Name, Department, Gender, and Marks.

The academic coordinator wants to identify students scoring above 80, compare department-wise performance, and determine

whether the data frame is suitable for further analysis.

Questions: a) Create a suitable data frame in R with at least six student records.

b) Use appropriate commands to inspect its structure and summary.

c) Filter students with Marks $\geq$ 80.

d) Evaluate whether the data frame is appropriate for student performance analysis. Justify your answer.

a) Create a suitable data frame in R with at least six student records.

Program:

```
# Create Student Data Frame
```

```
student <- data.frame(
```

```
  Student_ID = c(101,102,103,104,105,106),
```

```
  Name = c("Alice","Bob","Charlie","David","Eva","Frank"),
```

```
  Department = c("CSE","IT","ECE","IT","CSE","ECE"),
```

```
  Gender = c("Female","Male","Male","Male","Female","Male"),
```

```
  Marks = c(85,72,91,78,88,95)
```

```
)
```

Display the data frame

```
print(student)
```

output:

```
# Display the data frame
print(student)
Student_ID  Name Department Gender Marks
101      Alice      CSE Female    85
102       Bob       IT   Male    72
103  Charlie      ECE   Male    91
104   David       IT   Male    78
105    Eva      CSE Female    88
106   Frank      ECE   Male    95
```

b) Use appropriate commands to inspect its structure and summary.

Program:

```
# Display structure
```

```
str(student)
```

```
# Display summary
```

```
summary(student)
```

output:

```
> # Display structure
> str(student)
'data.frame': 6 obs. of 5 variables:
 $ Student_ID: num 101 102 103 104 105 106
 $ Name      : chr "Alice" "Bob" "Charlie" "David" ...
 $ Department: chr "CSE" "IT" "ECE" "IT" ...
 $ Gender    : chr "Female" "Male" "Male" "Male" ...
 $ Marks     : num 85 72 91 78 88 95
>
> # Display summary
> summary(student)
  Student_ID      Name      Department      Gender      Marks
Min.   :101.0  Length :6    Length :6    Length :6   Min.   :72.00
1st Qu.:102.2  N.unique :6    N.unique :3    N.unique :2  1st Qu.:79.75
Median :103.5  N.blank  :0    N.blank  :0    N.blank  :0   Median :86.50
Mean   :103.5  Min.nchar:3    Min.nchar:2   Min.nchar:4   Mean   :84.83
3rd Qu.:104.8  Max.nchar:7    Max.nchar:3    Max.nchar:6   3rd Qu.:90.25
Max.   :106.0                                     Max.   :95.00
```

c) Filter students with Marks > 80.

Program:

```
# Filter students with marks greater than 80
```

```
result <- subset(student, Marks > 80)
```

```
# Display result
```

```
print(result)
```

output:

```
> # Filter students with marks greater than 80
> result <- subset(student, Marks > 80)
>
> # Display result
> print(result)
```

	Student_ID	Name	Department	Gender	Marks
1	101	Alice	CSE	Female	85
3	103	Charlie	ECE	Male	91
5	105	Eva	CSE	Female	88
6	106	Frank	ECE	Male	95

d) Evaluate whether the data frame is appropriate for student performance analysis. Justify your answer.

Answer:

Yes, the data frame is appropriate for student performance analysis because:

- It stores all student details in a structured tabular format.
- Each column represents a specific attribute such as Student_ID, Name, Department, Gender, and Marks.
- It allows filtering, sorting, grouping, and summarizing student records easily.
- Department-wise and gender-wise performance analysis can be performed.
- Statistical measures such as average, maximum, minimum, and median marks can be calculated.
- The data frame can be used for visualization and further data analysis in R.

Conclusion:

The data frame is well-organized and suitable for analyzing student performance, generating reports, and supporting academic decision-making.

2. Case Study 2: Hospital Patient Records

Case Study: A hospital maintains a data frame with Patient_ID, Age, Gender, Disease, Admission_Days, and Treatment_Cost. Some records have missing treatment costs, and the administrator wants to identify patients with long hospital stays and evaluate the reliability of the data frame for cost analysis.

a) Create or assume a patient data frame.

Answer:

```
# Create Patient Data Frame
```

```
patient <- data.frame(  
  Patient_ID = c(201,202,203,204,205,206),  
  Age = c(45,60,35,50,28,65),  
  Gender = c("Male","Female","Male","Female","Male","Female"),  
  Disease = c("Fever","Diabetes","Cancer","Asthma","Typhoid","Heart Disease"),  
  Admission_Days = c(5,10,12,4,8,15),  
  Treatment_Cost = c(5000, NA, 25000, 7000, NA, 30000)  
)
```

```
# Display the data frame
```

```
print(patient)  
  
> # Create Patient Data Frame  
> patient <- data.frame(  
+   Patient_ID = c(201,202,203,204,205,206),  
+   Age = c(45,60,35,50,28,65),  
+   Gender = c("Male","Female","Male","Female","Male","Female"),  
+   Disease = c("Fever","Diabetes","Cancer","Asthma","Typhoid","Heart Disease"),  
+   Admission_Days = c(5,10,12,4,8,15),  
+   Treatment_Cost = c(5000, NA, 25000, 7000, NA, 30000)  
+ )  
>  
> # Display the data frame  
> print(patient)  
  Patient_ID Age Gender      Disease Admission_Days Treatment_Cost  
1         201  45  Male        Fever              5           5000  
2         202  60 Female    Diabetes             10              NA  
3         203  35  Male        Cancer             12          25000  
4         204  50 Female     Asthma              4            7000  
5         205  28  Male     Typhoid              8              NA  
6         206  65 Female Heart Disease          15          30000
```

b) Identify missing values in Treatment_Cost.

Answer:

```
# Identify missing values
```

```
is.na(patient$Treatment_Cost)
```

```
# Display rows with missing Treatment_Cost
```

```
patient[is.na(patient$Treatment_Cost), ]
```

```

> # Identify missing values
> is.na(patient$Treatment_Cost)
[1] FALSE TRUE FALSE FALSE TRUE FALSE
>
> # Display rows with missing Treatment_Cost
> patient[is.na(patient$Treatment_Cost), ]
  Patient_ID Age Gender Disease Admission_Days Treatment_Cost
2         202  60 Female Diabetes             10             NA
5         205  28  Male  Typhoid              8             NA

```

c) Filter patients with Admission_Days > 7.

Answer:

```
# Filter patients admitted for more than 7 days
```

```
result <- subset(patient, Admission_Days > 7)
```

```
# Display result
```

```
print(result)
```

```

> # Filter patients admitted for more than 7 days
> result <- subset(patient, Admission_Days > 7)
>
> # Display result
> print(result)
  Patient_ID Age Gender Disease Admission_Days Treatment_Cost
2         202  60 Female Diabetes             10             NA
3         203  35  Male  Cancer              12          25000
5         205  28  Male  Typhoid              8             NA
6         206  65 Female Heart Disease          15          30000

```

d) Evaluate whether missing values should be removed or replaced before calculating the average treatment cost.

Answer:

Missing values should be handled before calculating the average treatment cost because they can affect the accuracy of the result.

- If only a few values are missing, they can be **removed** using `na.omit()`.
- If many values are missing, they should be **replaced** with an appropriate value such as the **mean** or **median** treatment cost.
- Replacing missing values helps preserve the number of patient records and provides a more reliable analysis.
- Simply ignoring missing values may lead to inaccurate or biased conclusions.

Conclusion:

For hospital cost analysis, missing values should be cleaned by removing or replacing them before calculating the average treatment cost to ensure accurate and reliable results.

3. Case Study 3: Retail Sales Data Frame

Case Study: A retail store records Product_ID, Product_Name, Category, Quantity_Sold, Unit_Price, and

Sales_Date.

The manager wants to identify the best-selling products and create a new Total_Sales column.

Questions: a) Analyze the structure of the data frame.

b) Create Total_Sales using Quantity_Sold and Unit_Price.

c) Sort the data frame by Total_Sales in descending order.

d) Evaluate whether sorting alone is sufficient to identify the store's most successful products.

a)

Create Retail Sales Data Frame

```
sales <- data.frame(  
  Product_ID = c(101,102,103,104,105,106),  
  Product_Name = c("Laptop","Mobile","Headphones","Keyboard","Mouse","Monitor"),  
  Category =  
  c("Electronics","Electronics","Accessories","Accessories","Accessories","Electronics"),  
  Quantity_Sold = c(10,20,30,15,25,8),  
  Unit_Price = c(50000,20000,2000,1500,800,12000),  
  Sales_Date = as.Date(c("2025-01-10","2025-01-11","2025-01-12",  
    "2025-01-13","2025-01-14","2025-01-15"))  
)
```

Display structure

```
str(sales)
```

Display summary

```
summary(sales)
```

```
> # Display structure  
> str(sales)  
'data.frame': 6 obs. of 6 variables:  
 $ Product_ID : num 101 102 103 104 105 106  
 $ Product_Name : chr "Laptop" "Mobile" "Headphones" "Keyboard" ...  
 $ Category : chr "Electronics" "Electronics" "Accessories" "Accessories" ...  
 $ Quantity_Sold: num 10 20 30 15 25 8  
 $ Unit_Price : num 50000 20000 2000 1500 800 12000  
 $ Sales_Date : Date, format: "2025-01-10" "2025-01-11" "2025-01-12" "2025-01-13" ...  
>  
> # Display summary  
> summary(sales)  
 Product_ID Product_Name Category Quantity_Sold Unit_Price Sales_Date  
 Min. :101.0 Length : 6 Length : 6 Min. : 8.00 Min. : 800 Min. :2025-01-10  
 1st Qu.:102.2 N.unique : 6 N.unique : 2 1st Qu.:11.25 1st Qu.: 1625 1st Qu.:2025-01-11  
 Median :103.5 N.blank : 0 N.blank : 0 Median :17.50 Median : 7000 Median :2025-01-12  
 Mean :103.5 Min.nchar: 5 Min.nchar:11 Mean :18.00 Mean :14383 Mean :2025-01-12  
 3rd Qu.:104.8 Max.nchar:10 Max.nchar:11 3rd Qu.:23.75 3rd Qu.:18000 3rd Qu.:2025-01-13  
 Max. :106.0 Max. :30.00 Max. :50000 Max. :2025-01-15
```

b)

```
# Create Total_Sales column
sales$Total_Sales <- sales$Quantity_Sold * sales$Unit_Price
```

```
# Display updated data frame
print(sales)
```

```
> # Create Total_Sales column
> sales$Total_Sales <- sales$Quantity_Sold * sales$Unit_Price
>
> # Display updated data frame
> print(sales)
```

	Product_ID	Product_Name	Category	Quantity_Sold	Unit_Price	Sales_Date	Total_Sales
1	101	Laptop	Electronics	10	50000	2025-01-10	500000
2	102	Mobile	Electronics	20	20000	2025-01-11	400000
3	103	Headphones	Accessories	30	2000	2025-01-12	60000
4	104	Keyboard	Accessories	15	1500	2025-01-13	22500
5	105	Mouse	Accessories	25	800	2025-01-14	20000
5	106	Monitor	Electronics	8	12000	2025-01-15	96000

c)

```
# Sort by Total_Sales (Descending)
sorted_sales <- sales[order(-sales$Total_Sales), ]
```

```
# Display sorted data frame
print(sorted_sales)
```

```
# Sort by Total_Sales (Descending)
sorted_sales <- sales[order(-sales$Total_Sales), ]

# Display sorted data frame
print(sorted_sales)
```

	Product_ID	Product_Name	Category	Quantity_Sold	Unit_Price	Sales_Date	Total_Sales
	101	Laptop	Electronics	10	50000	2025-01-10	500000
	102	Mobile	Electronics	20	20000	2025-01-11	400000
	106	Monitor	Electronics	8	12000	2025-01-15	96000
	103	Headphones	Accessories	30	2000	2025-01-12	60000
	104	Keyboard	Accessories	15	1500	2025-01-13	22500
	105	Mouse	Accessories	25	800	2025-01-14	20000

d)

Evaluate whether sorting alone is sufficient to identify the store's most successful products.

Answer:

Sorting by **Total_Sales** helps identify products with the highest sales revenue, but it is **not sufficient** to determine the store's most successful products because:

- High revenue may result from a **high price**, not high demand.
- Quantity sold should also be analyzed.
- Product profit, customer demand, and sales trends should be considered.
- Category-wise performance and seasonal sales can provide better insights.

Conclusion:

Sorting by **Total_Sales** is useful, but additional analysis such as profit, quantity sold, and customer demand is required to accurately identify the store's most successful products.

4. Case Study 4: Employee Salary Analysis

Case Study: An organization stores employee records with Employee_ID, Name, Department, Experience, and Salary.

The HR manager wants to compare salaries across departments and identify employees whose salary is below the department average.

Questions: a) Select only the relevant columns for salary analysis.

b) Analyze how employees can be compared with their department averages.

c) Create a logical condition to identify employees below the overall average salary.

d) Evaluate the limitations of using only salary to assess employee performance.

a)

```
# Create Employee Data Frame
employee <- data.frame(
  Employee_ID = c(101,102,103,104,105,106),
  Name = c("Alice","Bob","Charlie","David","Eva","Frank"),
  Department = c("HR","IT","Finance","IT","HR","Finance"),
  Experience = c(3,5,7,2,4,6),
  Salary = c(35000,50000,60000,45000,40000,55000)
)

# Select relevant columns
salary_data <- employee[, c("Employee_ID", "Department", "Salary")]

# Display selected columns
print(salary_data)
```

Employee_ID	Department	Salary
101	HR	35000
102	IT	50000
103	Finance	60000
104	IT	45000
105	HR	40000
106	Finance	55000

b)

```
# Calculate department-wise average salary
dept_avg <- aggregate(Salary ~ Department, data = employee, mean)

# Display department averages
print(dept_avg)
```



```

· # Calculate department-wise average salary
· dept_avg <- aggregate(Salary ~ Department, data = employee, mean)
·
· # Display department averages
· print(dept_avg)
  Department Salary
·   Finance  57500
·         HR   37500
·         IT   47500

```

Analysis:

- Compare each employee's salary with the average salary of their department.
- Employees earning less than the department average can be identified for salary review.

c)

```

# Calculate overall average salary
avg_salary <- mean(employee$Salary)

# Display employees below average salary
result <- subset(employee, Salary < avg_salary)

print(result)

# Calculate overall average salary
avg_salary <- mean(employee$Salary)

# Display employees below average salary
result <- subset(employee, Salary < avg_salary)

print(result)
Employee_ID Name Department Experience Salary
      101 Alice           HR             3  35000
      104 David           IT             2  45000
      105 Eva            HR             4  40000

```

d) Evaluate the limitations of using only salary to assess employee performance.

Answer:

Using only **salary** to assess employee performance has several limitations:

- Salary may depend on **experience, job role, and seniority**.
- It does not measure **work quality or productivity**.
- Employee skills, teamwork, attendance, and performance ratings are not considered.
- Promotions and incentives may also affect salary.

Conclusion:

Salary alone is **not a reliable measure** of employee performance. A complete evaluation should include experience, performance ratings, productivity, skills, and other relevant factors.

5. Case Study 5: Environmental Pollution Monitoring

Case Study: An environmental agency collects City, Month, PM2.5, PM10, Temperature, and Air_Quality_Index in a data frame.

The agency wants to identify cities with dangerous air quality and evaluate whether the data frame can

support environmental decision-making.

Questions: a) Analyze the relationship between PM2.5 and Air_Quality_Index using the data frame.

b) Filter records where Air_Quality_Index exceeds a specified danger threshold.

c) Identify possible data-quality problems.

d) Evaluate whether the data frame contains enough information for long-term pollution policy decisions.

```
a)
# Create Environmental Data Frame
pollution <- data.frame(
  City = c("Chennai", "Delhi", "Mumbai", "Bengaluru", "Kolkata", "Hyderabad"),
  Month = c("Jan", "Jan", "Jan", "Jan", "Jan", "Jan"),
  PM2.5 = c(45, 120, 70, 40, 95, 55),
  PM10 = c(80, 180, 110, 70, 150, 90),
  Temperature = c(30, 18, 28, 25, 22, 29),
  Air_Quality_Index = c(90, 220, 140, 80, 180, 110)
)

# Analyze relationship
cor(pollution$PM2.5, pollution$Air_Quality_Index)

ouput:
[1] 0.99
```

Analysis:

- A high positive correlation (0.99) indicates that as **PM2.5 increases, the Air Quality Index (AQI) also increases**.
- Higher PM2.5 values generally indicate poorer air quality.

```
b)
# Filter cities with AQI greater than 150
danger <- subset(pollution, Air_Quality_Index > 150)

# Display result
print(danger)
```

	City	Month	PM2.5	PM10	Temperature	Air_Quality_Index
2	Delhi	Jan	120	180	18	220
5	Kolkata	Jan	95	150	22	180

c) Identify possible data-quality problems.

Answer:

Possible data-quality problems include:

- Missing values (NA) in pollution measurements.
- Duplicate records.
- Incorrect or unrealistic PM2.5, PM10, or AQI values.
- Inconsistent city or month names.
- Outdated or incomplete data.

d) Evaluate whether the data frame contains enough information for long-term pollution policy decisions.

Answer:

The data frame provides useful information about pollution levels, but it is **not sufficient** for long-term policy decisions because:

- It contains data for only a limited time period.
- Additional factors such as rainfall, wind speed, humidity, population, traffic, and industrial emissions are not included.
- Long-term historical data is needed to identify pollution trends.
- Geographic and seasonal variations should also be considered.

Conclusion:

The data frame is suitable for basic pollution analysis but requires additional environmental and historical data to support effective long-term pollution policy decisions.

6. Case Study 6: Online Course Enrollment Data

Case Study: An online learning platform stores Course_ID, Course_Name, Category, Enrolled_Students,

Completion_Rate, and Rating.

Management wants to identify courses with high enrollment but low completion rates.

Questions: a) Filter courses with Enrolled_Students > 1000 and Completion_Rate < 50.

- b) Analyze whether high enrollment indicates course success.
- c) Create a new performance category based on completion rate.
- d) Evaluate the usefulness of the data frame for improving course quality.

a)

```
# Create Online Course Data Frame
course <- data.frame(
  Course_ID = c(101,102,103,104,105,106),
  Course_Name = c("Python","R Programming","Data Science","AI Basics","Machine
Learning","Cyber Security"),
  Category = c("Programming","Programming","Data Science","AI","AI","Security"),
  Enrolled_Students = c(1500,900,2000,1200,800,1800),
  Completion_Rate = c(45,75,40,65,85,35),
  Rating = c(4.5,4.2,4.7,4.0,4.8,4.3)
)

# Filter courses
result <- subset(course,
  Enrolled_Students > 1000 &
  Completion_Rate < 50)

print(result)

# Filter courses
result <- subset(course,
  Enrolled_Students > 1000 &
  Completion_Rate < 50)

print(result)
```

Course_ID	Course_Name	Category	Enrolled_Students	Completion_Rate	Rating
101	Python	Programming	1500	45	4.5
103	Data Science	Data Science	2000	40	4.7
106	Cyber Security	Security	1800	35	4.3

b) Analyze whether high enrollment indicates course success.

Answer:

High enrollment **does not always indicate** course success because:

- Many students may enroll but fail to complete the course.
- Completion rate reflects learner engagement.
- Ratings and student feedback also measure course quality.
- A successful course should have high enrollment, high completion rate, and good ratings.

Conclusion:

High enrollment alone is **not sufficient** to determine course success.

c)

```
# Create Performance Category
```

```
course$Performance <- ifelse(course$Completion_Rate >= 80, "Excellent",  
                             ifelse(course$Completion_Rate >= 50, "Good", "Needs Improvement"))
```

```
# Display updated data frame
```

```
print(course)
```

```
# Display updated data frame  
print(course)
```

Course_ID	Course_Name	Category	Enrolled_Students	Completion_Rate	Rating	Performance
101	Python	Programming	1500	45	4.5	Needs Improvement
102	R Programming	Programming	900	75	4.2	Good
103	Data Science	Data Science	2000	40	4.7	Needs Improvement
104	AI Basics	AI	1200	65	4.0	Good
105	Machine Learning	AI	800	85	4.8	Excellent
106	Cyber Security	Security	1800	35	4.3	Needs Improvement

d) Evaluate the usefulness of the data frame for improving course quality.**Answer:**

The data frame is useful because it:

- Identifies courses with low completion rates.
- Compares course popularity using enrollment.
- Uses ratings to measure student satisfaction.
- Helps management identify courses that need improvement.

However, additional information such as **student feedback, instructor performance, course duration, and assessment results** would provide a better understanding of course quality.

Conclusion:

The data frame is useful for basic course analysis, but more detailed information is needed to make effective decisions for improving course quality.

7. Case Study 7: Banking Customer Data

Case Study: A bank maintains a data frame containing Customer_ID, Age, Account_Type, Balance,

Loan_Status, and Credit_Score.

The bank wants to identify customers who may be suitable for loan offers.

Questions: a) Select customers with Credit_Score > 700 and Balance > 50000.

b) Analyze whether these two conditions are sufficient for loan targeting.

c) Identify additional variables that should be included.

d) Evaluate the risk of making loan decisions using only the data frame variables given.

a)

Create Banking Customer Data Frame

```
customer <- data.frame(  
  Customer_ID = c(101,102,103,104,105,106),  
  Age = c(25,40,35,50,30,45),  
  Account_Type = c("Savings","Current","Savings","Current","Savings","Current"),  
  Balance = c(60000,45000,80000,70000,30000,90000),  
  Loan_Status = c("No","Yes","No","Yes","No","Yes"),  
  Credit_Score = c(720,680,750,710,690,780)  
)
```

Select eligible customers

```
result <- subset(customer,  
  Credit_Score > 700 &  
  Balance > 50000)
```

```
print(result)
```

```
print(result)  
Customer_ID Age Account_Type Balance Loan_Status Credit_Score  
101 25 Savings 60000 No 720  
103 35 Savings 80000 No 750  
104 50 Current 70000 Yes 710  
106 45 Current 90000 Yes 780  
.. ..
```

b) Analyze whether these two conditions are sufficient for loan targeting.

Answer:

No, these two conditions are **not sufficient** for loan targeting because:

- A high credit score shows good credit history.

- A high balance indicates financial stability.
- However, loan approval also depends on income, employment status, repayment history, and existing debts.

Conclusion:

Credit score and balance alone cannot accurately identify suitable loan customers.

c) Identify additional variables that should be included.

Answer:

Additional variables include:

- Monthly Income
- Employment Status
- Existing Loan Amount
- Repayment History
- Debt-to-Income Ratio
- Occupation
- Loan Purpose
- Customer Transaction History

These variables improve the accuracy of loan eligibility analysis.

d) Evaluate the risk of making loan decisions using only the data frame variables given.

Answer:

Using only the given variables has several risks:

- It may approve customers who cannot repay the loan.
- It ignores income and employment details.
- Previous repayment behavior is not fully considered.
- Financial obligations and debt are not included.
- This may increase the risk of loan defaults.

Conclusion:

The given data frame provides basic customer information, but additional financial and credit-related variables are required for accurate and reliable loan decision-making.

8. Case Study 8: Transportation Accident Records

Case Study: A transportation department stores Accident_ID, Location, Vehicle_Type, Weather, Severity,

and Number_of_Vehicles.

Officials want to analyze severe accidents and determine whether the data frame can be used to prioritize

road safety improvements.

Questions: a) Filter accidents classified as severe.

b) Analyze accident frequency by weather condition.

c) Create a summary of accidents by Vehicle_Type.

d) Evaluate whether accident frequency alone should determine road improvement priorities.

a)

Create Accident Data Frame

```
accident <- data.frame(  
  Accident_ID = c(101,102,103,104,105,106),  
  Location = c("Chennai","Delhi","Mumbai","Bengaluru","Hyderabad","Kolkata"),  
  Vehicle_Type = c("Car","Bus","Bike","Car","Truck","Bike"),  
  Weather = c("Rainy","Foggy","Sunny","Rainy","Foggy","Sunny"),  
  Severity = c("Severe","Minor","Severe","Moderate","Severe","Minor"),  
  Number_of_Vehicles = c(3,2,2,4,5,1)  
)
```

Filter severe accidents

```
result <- subset(accident, Severity == "Severe")
```

```
print(result)
```

```
print(result)  
Accident_ID Location Vehicle_Type weather Severity Number_of_vehicles  
      101    Chennai         Car   Rainy   Severe             3  
      103     Mumbai         Bike   Sunny   Severe             2  
      105  Hyderabad         Truck  Foggy   Severe             5  
" - - - - -"
```

b) Analyze accident frequency by weather condition.

Answer:

```
# Count accidents by weather
```

```
table(accident$Weather)
```

Output:

```
Foggy Rainy Sunny
```

```
2 2 2
```

Analysis:

- Rainy weather: 2 accidents
- Foggy weather: 2 accidents
- Sunny weather: 2 accidents

This helps identify weather conditions associated with accident occurrences.

c) Create a summary of accidents by Vehicle_Type.

Answer:

```
# Summary by Vehicle Type
```

```
table(accident$Vehicle_Type)
```

Output:

```
Bike Bus Car Truck
```

```
2 1 2 1
```

d) Evaluate whether accident frequency alone should determine road improvement priorities.

Answer:

Accident frequency **alone is not sufficient** to determine road improvement priorities because:

- Accident severity should also be considered.
- Road conditions and infrastructure affect accident rates.
- Weather conditions and traffic volume influence accidents.
- Causes such as speeding, poor lighting, and driver behavior should be analyzed.

Conclusion:

Road safety improvements should be based on **accident frequency, severity, road conditions, traffic patterns, and environmental factors**, rather than accident frequency alone.

9. Case Study 9: E-Commerce Customer Orders

Case Study: An e-commerce company stores Order_ID, Customer_ID, Product_Category, Order_Value,

Delivery_Days, and Customer_Rating.

The company wants to find whether delayed deliveries are associated with lower customer ratings.

Questions: a) Create a logical condition for delayed orders.

b) Analyze the ratings of delayed and non-delayed orders.

c) Identify whether the data frame supports a meaningful business conclusion.

d) Evaluate possible confounding factors that may affect customer ratings.

a) Create a logical condition for delayed orders.

```
orders$Delayed <- orders$Delivery_Days > 5
```

b) Analyze the ratings of delayed and non-delayed orders.

```
aggregate(Customer_Rating ~ Delayed, data = orders, mean)
```

c) Identify whether the data frame supports a meaningful business conclusion.

Answer: Yes. The data frame helps compare customer ratings for delayed and non-delayed orders to identify any relationship.

d) Evaluate possible confounding factors that may affect customer ratings.

Answer: Product quality, customer expectations, order value, delivery service, and customer support may also affect ratings.

10. Case Study 10: Agricultural Crop Yield Data

Case Study: An agricultural research center stores Crop, Soil_Type, Rainfall, Fertilizer_Used,

Temperature, and Yield in a data frame.

Researchers want to compare crop yields and determine which factors may be associated with higher productivity.

Questions: a) Analyze the data frame using summary() and appropriate filtering.

b) Compare yield values for different crops or soil types.

c) Identify potential outliers in Yield.

d) Evaluate whether the data frame is sufficient to recommend a single fertilizer strategy for all farms.

a) Analyze the data frame using summary() and appropriate filtering.

```
# Create data frame
```

```
crop <- data.frame(
```

```

Crop = c("Rice", "Wheat", "Rice", "Maize", "Wheat"),
Soil_Type = c("Clay", "Loam", "Clay", "Sandy", "Loam"),
Rainfall = c(200, 150, 220, 100, 160),
Fertilizer_Used = c("A", "B", "A", "C", "B"),
Temperature = c(28, 25, 30, 32, 26),
Yield = c(5.2, 4.5, 5.8, 3.9, 4.8)
)

```

```
# Summary
```

```
summary(crop)
```

```
# Filter Yield > 5
```

```
subset(crop, Yield > 5)
```

Answer: summary() gives statistical details of the data. Filtering displays crops with **Yield greater than 5**.

b) Compare yield values for different crops or soil types.

```
aggregate(Yield ~ Crop, data = crop, mean)
```

```
aggregate(Yield ~ Soil_Type, data = crop, mean)
```

Answer: Compare the **average yield** of each crop and soil type to identify higher productivity.

c) Identify potential outliers in Yield.

```
boxplot(crop$Yield, main = "Yield Boxplot")
```

Answer: The boxplot helps identify **unusually high or low yield values (outliers)**.

d) Evaluate whether the data frame is sufficient to recommend a single fertilizer strategy for all farms.

Answer: No. The data frame alone is **not sufficient** because crop yield also depends on **soil type, rainfall, temperature, crop variety, and other environmental factors**. More data and analysis are needed before recommending a single fertilizer strategy for all farms.